

High-Dimensional Deterministic Chaos in Solar Metric-Radio Spikes Event

Adolfo L. Mendez¹ and Mayte Alvarez¹

¹Math and Natural Science Department, Miami Dade College, Padron Campus, Miami, FL

Faculty Mentor: *Adolfo L. Mendez, PhD, Mathematics and Natural Sciences, Padron Campus, Miami Dade College*

Abstract

Solar radio emission associated with Type II, III, and IV events often displays chains or groups with fine temporal structures. The richness of these fine structures reflects the underlying plasma trigger mechanisms and scenarios at different levels of the magnetic structure hosting the sources of such events in the solar corona. These fine structures have been classified, primarily based on their multi-frequency dynamical spectral profiles, into four fundamental groups: periodic or quasi-periodic pulsations, intermediate drift bursts (fibers), zebra patterns, and spike events, among others. The richness of events is the clue for understanding some kind of fragmentation processes during the acceleration, trapping, and instabilities of the released energy during the main phase of solar flares. In this contribution, we analyze the high-resolution time series data (1 ms) associated with a group of spike events recorded at three fixed metric frequencies by the Basovizza Observing Station (Trieste Astronomical Observatory). This work continues previous studies in the characterization of the dynamics of these fine temporal structures to contribute to clarifying the fundamental differences between them.

Significance

Fine temporal structures in solar radio emissions serve as powerful diagnostics of the plasma processes governing energy release in the solar corona. By applying nonlinear time-series methods with millisecond resolution, this work explores whether spike events exhibit high-dimensional deterministic chaos rather than random fluctuations. Detecting deterministic patterns would provide evidence of self-organized plasma instabilities and fragmentation during solar flares, improving our understanding of acceleration and trapping mechanisms. These findings contribute to refining coronal models and enhancing the interpretation of radio diagnostics relevant to space weather forecasting.

Keywords

Solar radio bursts; spike events; metric emission; nonlinear time series; deterministic chaos.